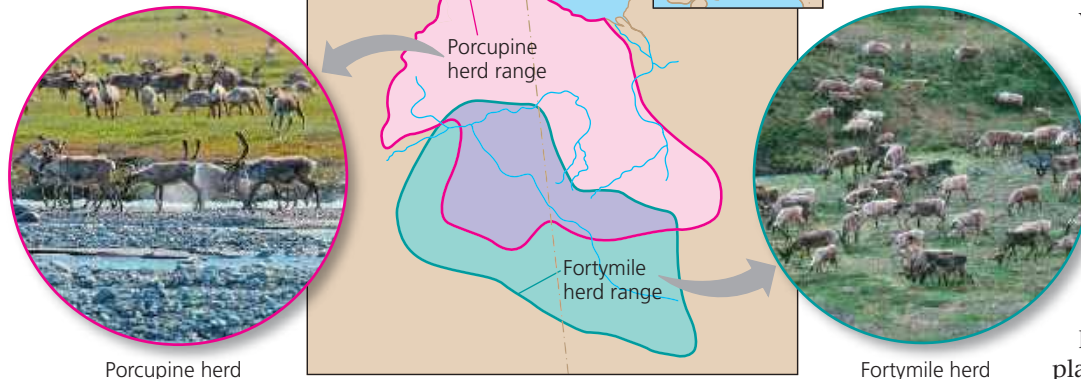


▼ **Figure 23.6 One species, two populations.** These two caribou populations in the Yukon are not totally isolated; they sometimes share the same area (shaded purple). Still, members of each population are most likely to breed within their own population.



one or more factors must be at work. In this section, we'll explore one way to test whether evolution is occurring in a population.

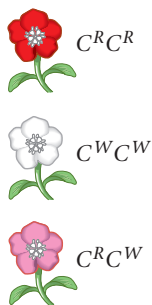
Gene Pools and Allele Frequencies

A **population** is a group of individuals of the same species that live in the same area and interbreed, producing fertile offspring. Different populations of a species may be isolated geographically from one another, exchanging genetic material only rarely. This is common for species that live on widely separated islands or in different lakes, but not all populations are isolated (**Figure 23.6**). Still, members of a population typically breed with one another and thus are more closely related to each other than to members of other populations.

We can characterize a population's genetic makeup by describing its **gene pool**, which consists of all copies of every type of allele at every locus in all members of the population. If only one allele exists for a particular locus in a population, that allele is said to be *fixed* in the gene pool, and all individuals are homozygous for that allele. But if there are two or more alleles for a particular locus in a population, individuals may be either homozygous or heterozygous.

For example, imagine a population of 500 wildflower plants with two alleles, C^R and C^W , for a locus that codes for flower pigment. These alleles show incomplete dominance; thus, each genotype has a distinct phenotype. Plants homozygous for the C^R allele ($C^R C^R$) produce red pigment and have red flowers; plants homozygous for the C^W allele ($C^W C^W$) produce no red pigment and have white flowers; and heterozygotes ($C^R C^W$) produce some red pigment and have pink flowers.

Each genotype and each allele has a frequency (proportion) in the population. To see how these frequencies are calculated, suppose our population has 320 plants with red flowers (genotype $C^R C^R$), 160 with pink flowers (genotype $C^R C^W$), and 20 with



white flowers (genotype $C^W C^W$). Since 320 of the 500 plants have red flowers, the frequency of the $C^R C^R$ genotype is 0.64 (320/500). Likewise, the frequency of the $C^R C^W$ genotype is 0.32 (160/500) and the frequency of the $C^W C^W$ genotype is 0.04 (20/500).

With respect to allele frequencies, wildflowers are diploid organisms, so each individual has two alleles for each of its genes. Hence, the 500 individuals in our population have a total of 1,000 copies of the gene for flower color. The C^R allele accounts for 800 of these copies ($320 \times 2 = 640$ for $C^R C^R$ plants, plus $160 \times 1 = 160$ for $C^R C^W$ plants). Thus, the frequency of the C^R allele is $800/1,000 = 0.8$ (80%).

When studying a locus with two alleles, the convention is to use p to represent the frequency of one allele and q to represent the frequency of the other allele. Thus, p , the frequency of the C^R allele in the gene pool of this population, is $p = 0.8$ (80%). And because there are only two alleles for this gene, the frequency of the C^W allele, represented by q , must be $q = 1 - p = 0.2$ (20%). For loci that have more than two alleles, the sum of all allele frequencies must still equal 1 (100%).

The Hardy-Weinberg Equation

One way to assess whether natural selection or other factors are causing evolution at a particular locus is to determine what the genetic makeup of a population would be if it were *not* evolving at that locus. We can then compare that with the data we actually observed for the population. If there are no differences, this suggests that the population is not evolving. If there are differences, this suggests that the population is evolving—and then we can investigate why.

Hardy-Weinberg Equilibrium

In a population that is not evolving, allele and genotype frequencies will remain constant from generation to generation, provided that only Mendelian segregation and recombination of alleles are at work. Such a population is said to be in **Hardy-Weinberg equilibrium**, named for the British mathematician and German physician, respectively, who independently developed this idea in 1908.

To determine whether a population is in Hardy-Weinberg equilibrium, it is helpful to think about genetic crosses in a new way. Previously, we used Punnett squares to determine the genotypes of offspring in a genetic cross (see Figure 14.5). Here, instead of considering the possible allele combinations from one cross, we'll consider the combination of alleles in *all* of the crosses in a population. Imagine that all the alleles for a given locus from all the individuals in a population are placed